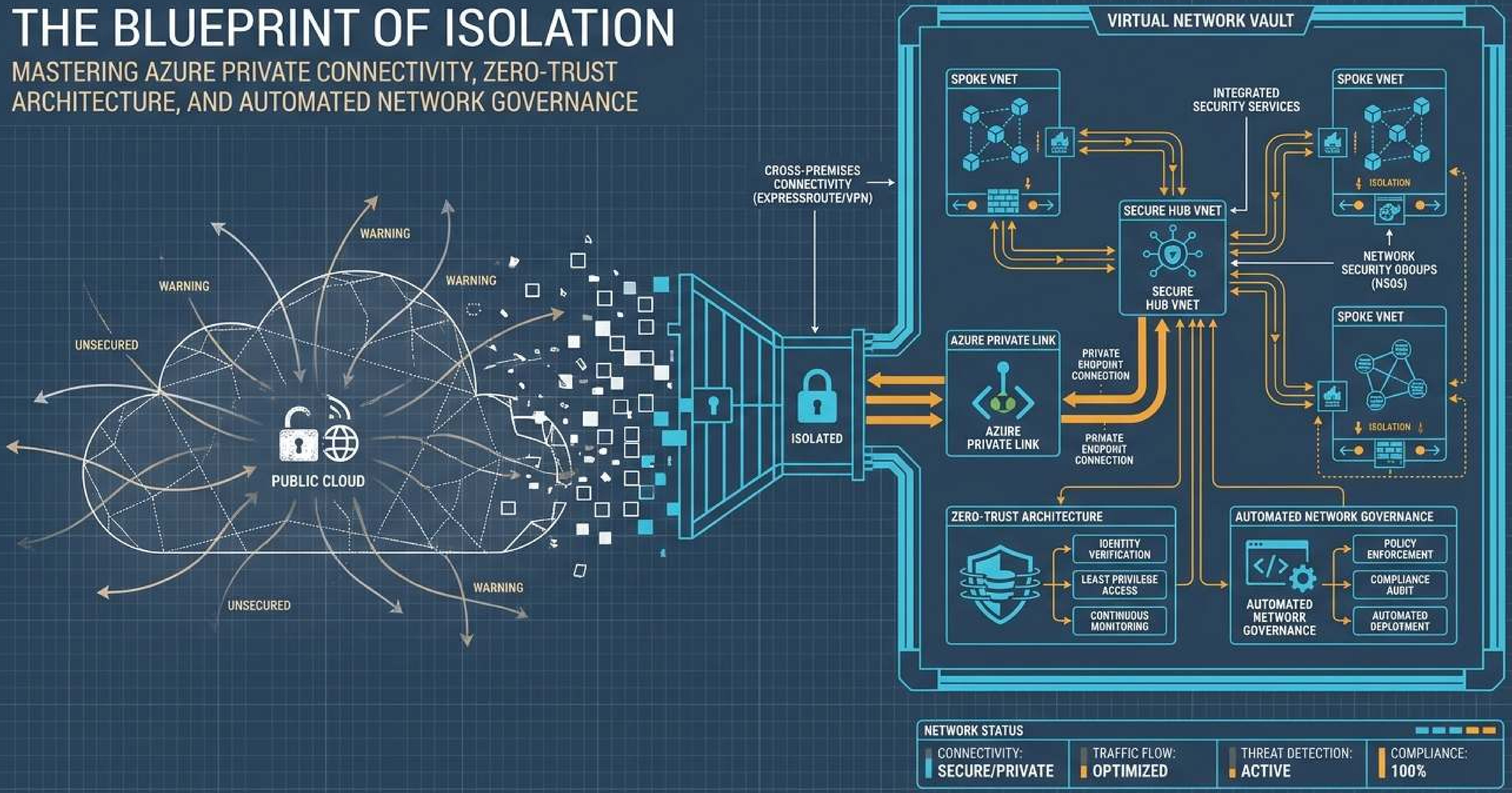
THE BLUEPRINT OF ISOLATION

MASTERING AZURE PRIVATE CONNECTIVITY, ZERO-TRUST ARCHITECTURE, AND AUTOMATED NETWORK GOVERNANCE



AUTHENTICATION IS ONLY HALF THE PERIMETER

THE DEFAULT STATE

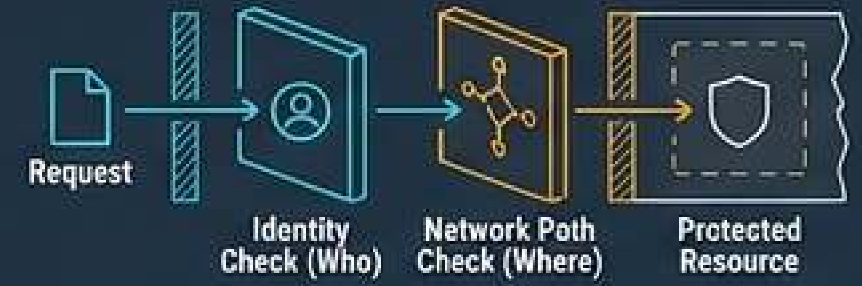
PaaS resources (SQL, Storage, Key Vault, OpenAI) are assigned internet-resolvable public endpoints upon creation.

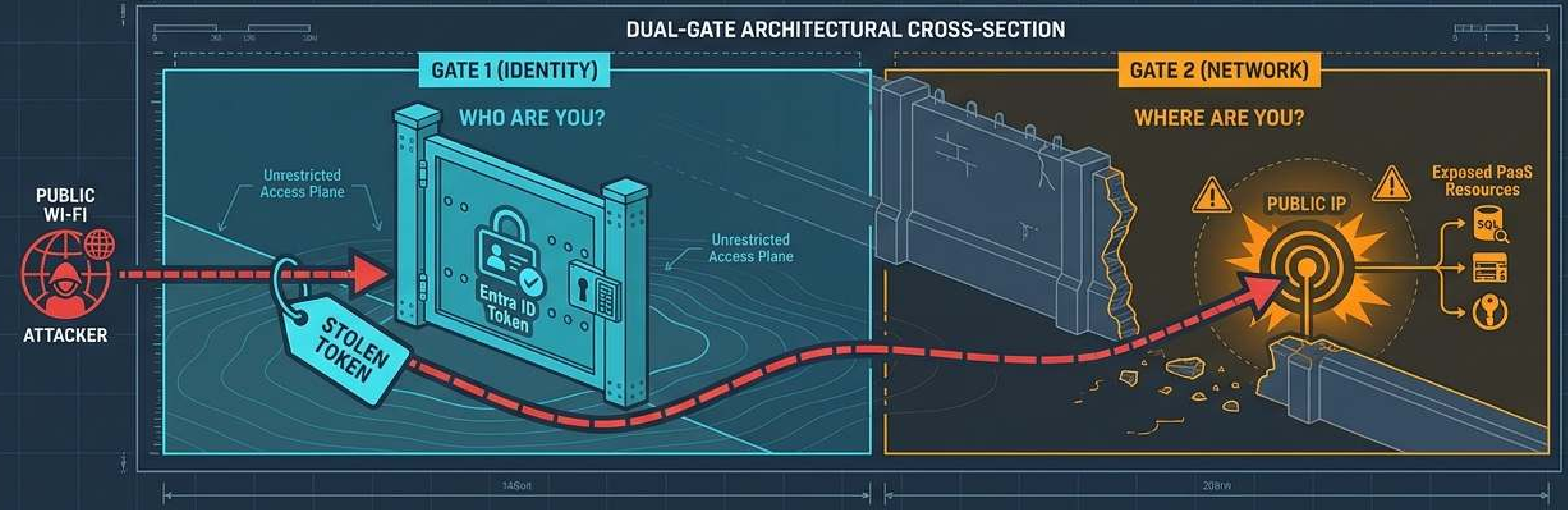
THE THREAT

A leaked credential or stolen token is valid from anywhere. Identity answers "who," but network controls dictate "from where."

THE REQUIREMENT

True defense-in-depth requires checking both the identity of the caller and the network path of the request.





SERVICE ENDPOINTS OPTIMIZE ROUTING BUT LEAVE THE DOOR AJAR

MECHANISM

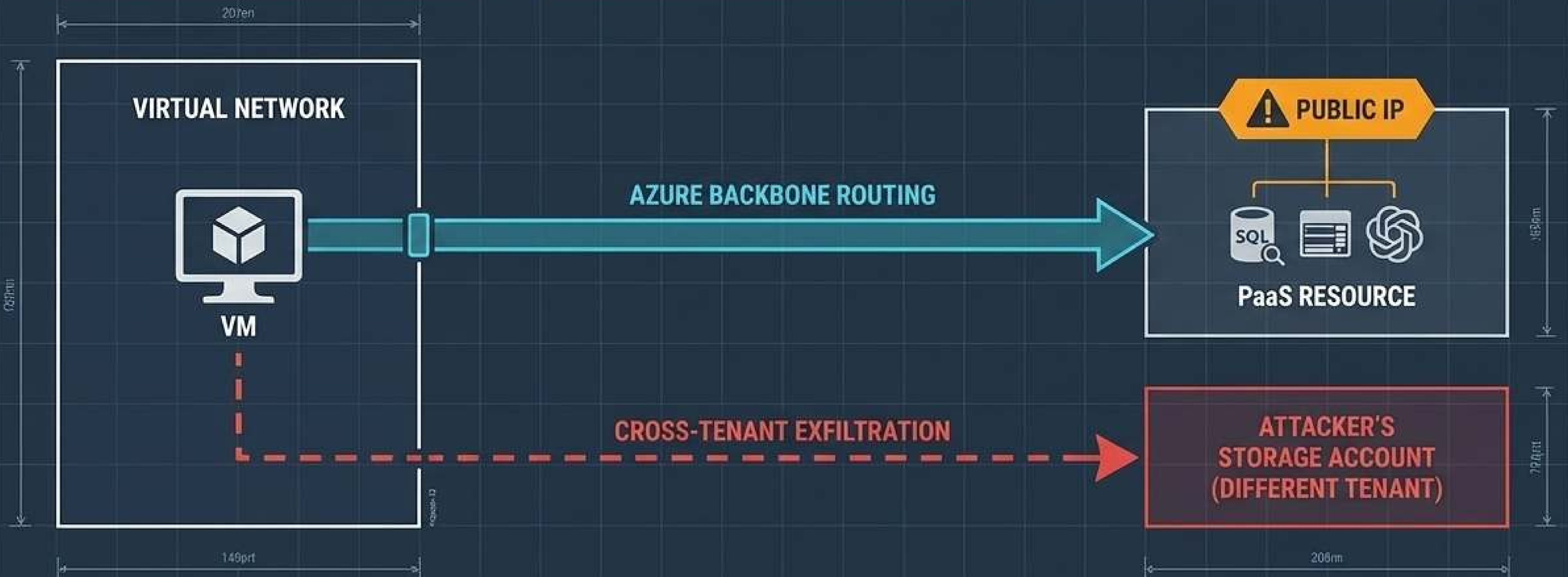
Extends VNet identity to the service, routing traffic over the Microsoft backbone instead of the public internet.

SCOPE LIMITATIONS

Grants access ‘service-wide’ (e.g., all of Azure Storage), not to a specific instance.

LINGERING RISKS

The FQDN still resolves to a public IP. External attackers can hit the public endpoint directly, and internal VMs can exfiltrate data to external tenants.



Private Endpoints Project PaaS Directly Into Your Virtual Network

THE ARCHITECTURE

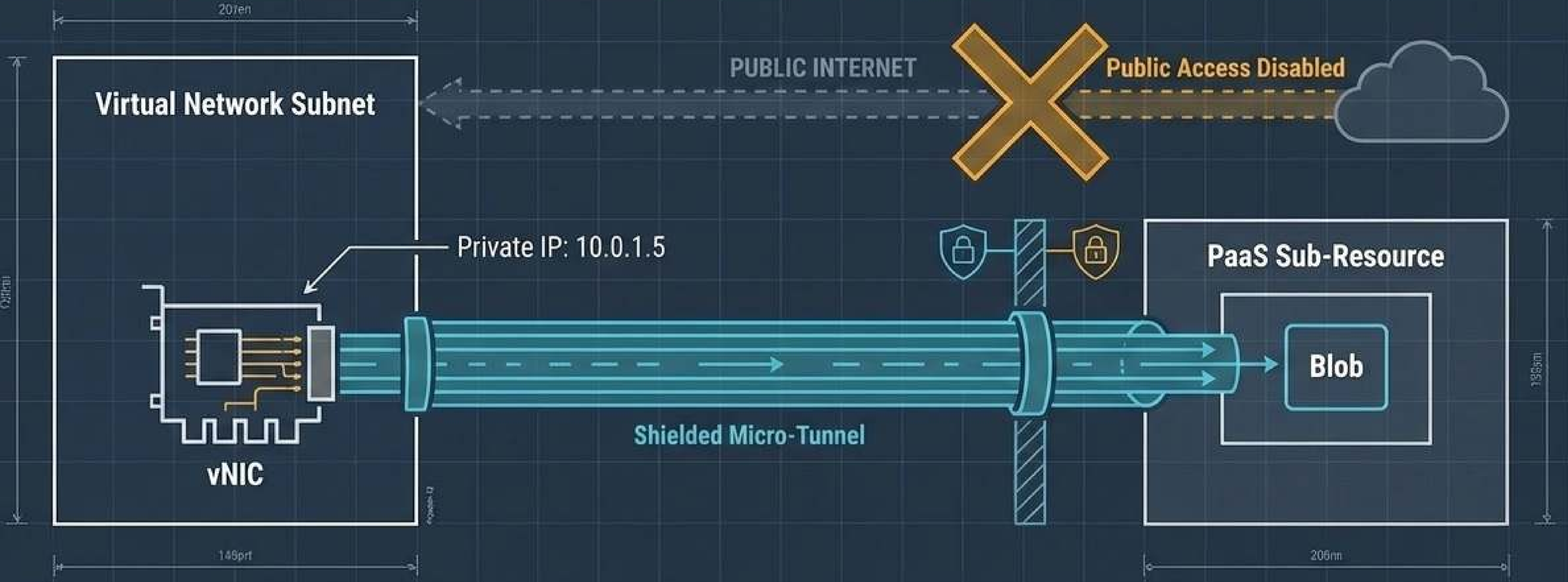
Azure deploys a physical vNIC directly into your subnet, assigning the PaaS resource an IP from your own private address space.

THE SCOPE

Connections are strictly 'instance-specific' (reaching only your specific database or vault), neutralizing cross-tenant exfiltration.

THE KILL SWITCH

Administrators can fully disable the public endpoint, ensuring stolen credentials are fundamentally useless outside the VNet.



COMPARING SERVICE AND PRIVATE ENDPOINTS

CRITERIA	SERVICE ENDPOINTS	PRIVATE ENDPOINTS
IP ADDRESSING	Public IP remains	Private IP (vNIC injected into VNet)
DNS RESOLUTION	Resolves to Public IP	Resolves to Private IP via Internal Zone
PUBLIC EXPOSURE	Always enabled	Can be permanently disabled
EXFILTRATION RISK	High (Cross-tenant paths open)	Low (Strictly instance-scoped)
COST & COMPLEXITY	Free, simpler setup	Hourly + data processing costs, requires DNS

DNS Integration Dictates the Success of the Private Path

The Trap

Creating a Private Endpoint NIC is insufficient. Without private DNS, traffic still attempts the public path and fails.

The Override

Azure uses standardized privatelink zones to hijack public FQDN resolution strictly within the VNet.

Complexity Alert

Multi-endpoint services require multiple zones. Azure OpenAI requires three simultaneous zones to function: `privatelink.cognitiveservices`, `privatelink.openai`, and `privatelink.services.ai`.

⚠ Without the DNS zone, the query bypasses the vNIC and hits a dead-end at the disabled public endpoint.

The Four Mechanics of a Private Link Connection

1. The NIC

The physical footprint in your subnet, drawing a private IP from your RFC 1918 address space.

2. The Sub-Resource

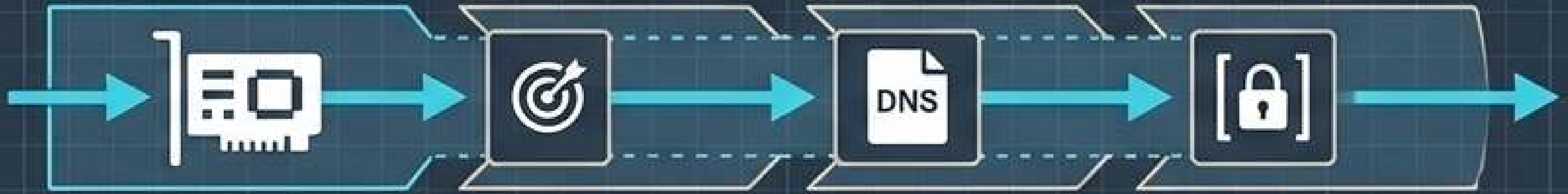
The specific target specification for services with multiple endpoints.

3. The DNS Record

The critical A-Record mapping the public FQDN to the new private IP via the privatelink zone.

4. The Approval Workflow

The trust mechanism. Auto-approved if requester and resource share an owner; manually gated for cross-tenant requests.



Private Link Service Reverses the Consumption Pattern

The **Shift:** Move from consuming Microsoft's platform to securely exposing your own internal APIs or multi-tenant SaaS offerings.

The Challenge

Exposing internal services to other VNets or distinct Entra tenants without issuing a public IP or dealing with complex network peering.

The Alias

Instead of sharing routing tables, providers share an opaque 'alias' string. Consumers use this string to request a private endpoint, keeping backend topologies completely hidden.

The Shift in Action

The Internal Mechanics of Provider-Side Exposure

Internal Load Balancer (ILB)

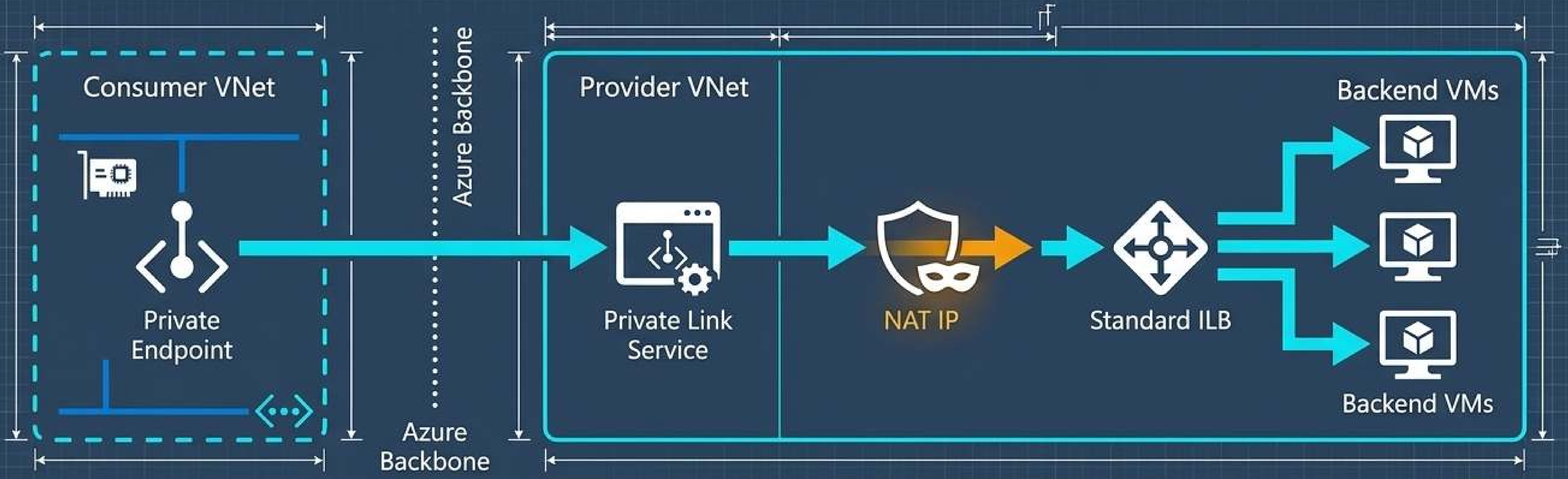
A Standard SKU ILB is strictly required as the frontend distributor for incoming Private Link requests.

Source NAT (SNAT)

The NAT IP masks the consumer's real IP, sidestepping overlapping IP conflicts entirely. The backend only ever sees the NAT IP.

One-Way Micro-Tunnel

Consumers can reach the ILB, but providers cannot initiate connections back to the consumer. The blast radius is restricted to the exposed service.



Service Exposure vs. Network Peering

Blast Radius Analysis: Comparing VNet Peering and Private Link Service for secure cross-network communication.

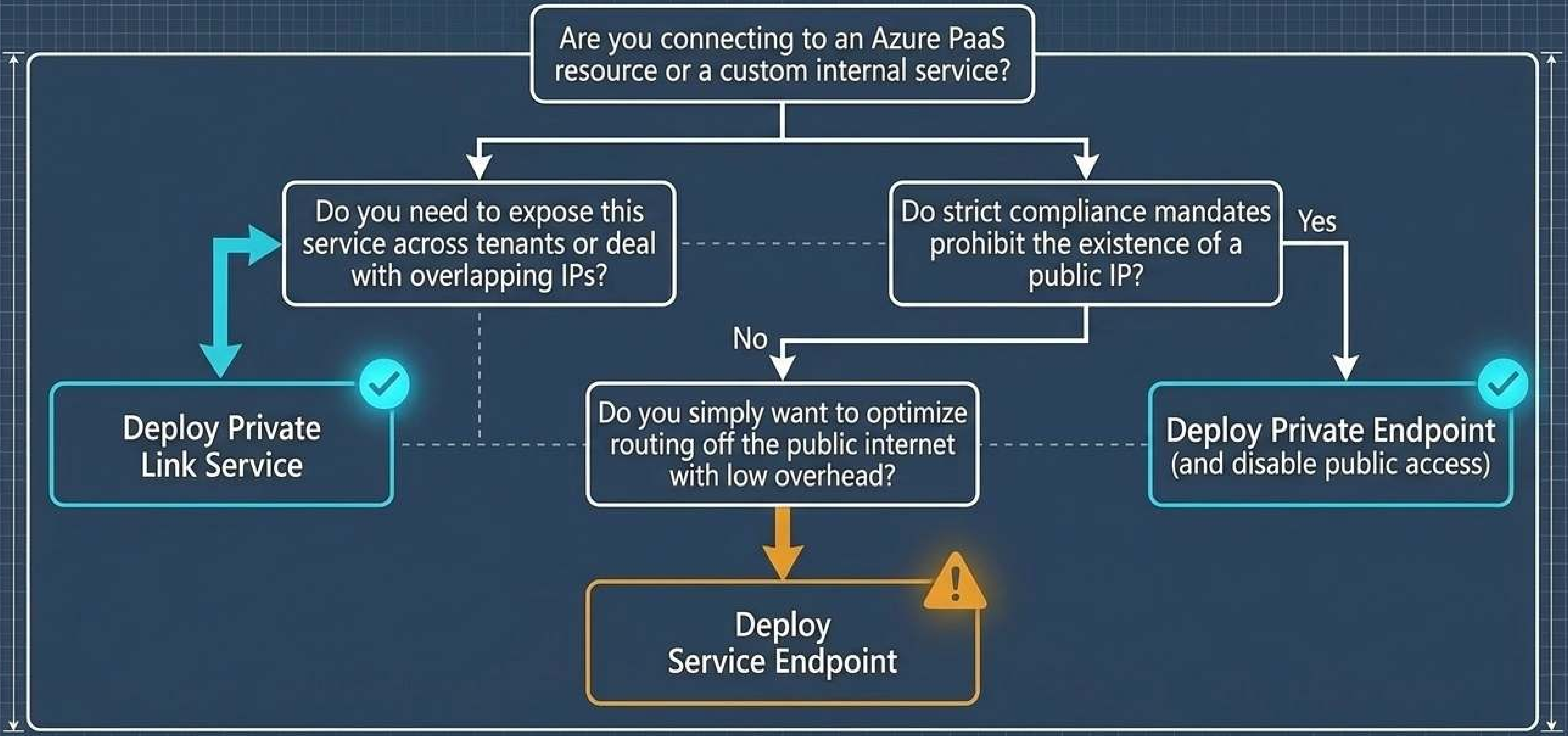
VNet Peering

- Full bidirectional network visibility.
- Fails instantly if IP address ranges overlap.
- Not natively supported across different Microsoft Entra tenants.

Private Link Service

- One-directional, strictly service-scoped.
- SNAT makes it completely immune to overlapping IP conflicts.
- Built natively to support cross-tenant and SaaS delivery architectures.

The Connectivity Diagnostic Tree



Automating the Baseline with Azure Policy

The Drift Problem

Without enforcement, developers provisioning new storage or databases will inherit the platform default: exposed public endpoints.

Detection

Deploy built-in policies using AuditIfNotExists to surface PaaS resources missing private endpoints.

Enforcement

Evaluate the publicNetworkAccess property. Apply a Deny effect aggressively block the creation of publicly exposed PaaS resources at deployment time.

Rollout Strategy

Audit everywhere → Deny in non-production → Deny in production

Validating the Perimeter with Defender for Cloud

Unified Signal

Defender continuously scans the environment, surfacing missing private endpoints as high-priority networking recommendations.

Attack Path Analysis

Defender correlates network exposure with identity and data. A public endpoint on a database with sensitive data creates a critical attack path; deploying a private endpoint severs that path.

Closed-Loop Remediation

Resolving a Defender recommendation by deploying a private endpoint simultaneously satisfies Azure Policy compliance checks.

The Blueprint of Zero-Trust Network Architecture

Identity + Network

Authentication proves identity; Private Endpoints prove location.

Both are mandatory.

Precision Exposure

Private Link Service enables robust cross-tenant API sharing without expanding the blast radius or exposing network topologies.

Continuous Enforcement

Secure baselines require automated governance. Trust Azure Policy to enforce the defaults, and Defender for Cloud to validate the perimeter.